Project and Data Management (PDM) Plan

PROJECT OVERVIEW

PROJECT TITLE:

SOLAR ENERGY PRODUCTION FORECASTING USING MACHINE LEARNING AND METEOROLOGICAL DATA

SUMMARY OF THE PROJECT TOPIC AND BACKGROUND:

While solar energy is an important part of the transition to renewable energy sources, it is highly variable based on weather changes during the day. Forecasting Solar Power Production Solar power has become a popular source of sustainable energy, and accurately forecasting its production

This project will be focused on developing machine learning models for predicting solar energy output based on historical solar power generation data, as well as meteorological parameters. Using historical trends and weather conditions like solar irradiance, temperature, and humidity, the project will produce short-term energy forecasts, to help solar farm operators and energy planners.

RESEARCH QUESTION:

How accurately can machine learning models predict solar energy production based on historical meteorological and solar power generation data?

PROJECT OBJECTIVES:

Data Collection & Preprocessing – Obtain Solar power generation and weather data from credible sources, and clean and preprocess the data for further analysis.

EDA – Explore Trends, correlation and patterns in between Meteorological variables and Solar energy output.

Feature Engineering – Extract and select the most relevant features impacting solar power production — this may include solar irradiance, temperature, humidity and wind speed.

Model Implementation & Comparison – Use multiple machine-learning models (Random Forest, XGBoost, LSTMs) to predict solar energy output.

Model Validation & Optimization — Assess model performance using the RMSE, MAE and R^2 metrics, and optimize hyperparameters to improve accuracy**.**

Deploy & Interpret– Appropriately deploy the best model, interpret results, and deliver insights into how to use these predictions for real energy management in the field.

REFERENCE LIST

1. Antonanzas, J., Osorio, N., Escobar, R., Urraca, R., Martinez-de-Pison, F. and Antonanzas-Torres, F., 2016. Review of photovoltaic power forecasting. *Solar Energy*, 136, pp. 78-111. Available at: https://doi.org/10.1016/j.solener.2016.06.069 [Accessed 10 February 2025].

2. Voyant, C., Notton, G., Kalogirou, S., Nivet, M.L., Paoli, C., Motte, F. and Fouilloy, A., 2017. Machine learning methods for solar radiation forecasting: A review. *Renewable Energy*, 105, pp. 569-582. Available at: https://doi.org/10.1016/j.renene.2016.12.095 [Accessed 10 February 2025].

3. Wan, C., Zhao, J., Song, Y., Xu, Z., Lin, J. and Hu, Z., 2015. Photovoltaic and solar power forecasting using machine learning: A state-of-the-art review. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 3(4), pp. 1474-1486. Available at: https://doi.org/10.1109/JESTPE.2015.2427378 [Accessed 10 February 2025].

PROJECT PLAN:

TASK LIST AND TIMELINE

Task	Description	Start Date	End Date
Project Proposal & Research	Define project scope, research relevant literature, and finalize research question.	Week 1	Week 2
Data Collection	Gather solar power generation and meteorological datasets from sources such as NREL, Kaggle, and Open-Meteo API.	Week 2	Week 3
Data Preprocessing & Cleaning	Handle missing values, format data correctly, and normalize variables for analysis.	Week 3	Week 4
Exploratory Data Analysis (EDA)	Visualize and analyze relationships between meteorological factors and solar power output.	Week 4	Week 5
Feature Engineering	Select key features (solar irradiance, temperature, humidity, wind speed) that impact energy production.	Week 5	Week 6
Model Development	Train and test various machine learning models (Random Forest, XGBoost, LSTMs) for prediction.	Week 6	Week 8
Model Evaluation & Optimization	Use RMSE, MAE, and R^2 metrics to validate models and fine-tune hyperparameters.	Week 8	Week 9
Final Model Selection & Interpretation	Choose the best-performing model, analyze results, and prepare deployment strategy.	Week 9	Week 10
Project Report Draft	Write initial project report with methodology, findings, and conclusions.	Week 9	Week 10
Final Report & Presentation	Refine final report, create PowerPoint slides, and practice for presentation.	Week 10	Week 11

TIMELINE:

Week	Task
1-2	Project Proposal & Literature Review
2-3	Data Collection & PDM Plan Submission
3-4	Data Cleaning & Preprocessing
4-5	Exploratory Data Analysis (EDA)
5-6	Feature Engineering
6-7	Model Development
7-8	Model Evaluation & Optimization
8-9	Final Model Selection & Report Draft
9-10	Report Finalization & Presentation Preparation

Data Management Plan

Overview of the Dataset

- **Dataset:** https://www.kaggle.com/datasets/anikannal/solar-power-generation-data
- **Source:** Kaggle
- **Geographical Coverage:** Includes solar farms from multiple regions.

Data Collection

- **Primary Source:** Kaggle dataset.
- **Data Retrieval Format:** CSV files from Kaggle

Metadata

- **File Format:** CSV
- **Number of Records:** ~300,000 + data points
- **Expected File Size:** ~500MB-1GB after preprocessing
- **Key Variables:**
    - **DATE_TIME:** Timestamp of energy production
    - **SOLAR_IRRADIANCE (W/m²):** Amount of solar energy received
    - **TEMPERATURE (°C):** Ambient temperature
    - **HUMIDITY (%):** Moisture content in the air
    - **WIND SPEED (m/s):** Wind impact on solar efficiency
    - **ENERGY_OUTPUT (kW):** Power generated

Security & Storage

- **Primary Storage:** GitHub repository for code.
- **Backup Storage:** OneDrive or Google Drive for dataset backups.
- **Backup Frequency:** Weekly updates and data storage backups.

Ethical Requirements

- • **GDPR Compliance:** The dataset does not include any personal or sensitive data, it is all publicly available data.
- • **UH Ethical Policies**: Project adheres to University of Hertfordshire ethical research policies.
- • **Data Use Permission:** Open-Source Dataset for academic research
- • **Ethical Data Collection**: All data was initially sourced from reputable agencies and collected under regular research guideline